Lecture 3

Linear representations

Symmetry enters physics through linear spaces: in classical physics small oscillations add, and in quantum physics states superpose. This lecture sets up the central notion of the course — a linear representation of a group — together with the construction by which symmetries act on functions, and says when two representations are the same. Its last example, the rotations of a triangle, splits into two smaller representations over $\mathbb{C}$ but not over $\mathbb{R}$; splitting representations is the subject of Lecture 4.

By the end of this lecture you will be able to:

- Write down representations of a group, including permutation representations.
- Let a group act on functions by $\psi(g^{-1}x)$.
- Decide whether two representations are equivalent.

3.1 Representations: definition and first examples

Lectures 1 and 2 studied groups in the abstract. Physics, however, meets a group through its *action on the states of a system*, and the state spaces of physics — quantum Hilbert spaces, spaces of displacements of atoms, spaces of fields — are vector spaces. The operations of the group must respect that linear structure: a symmetry operation applied to a superposition is the superposition of the transformed states. The mathematical object capturing this is the following.

Definition 3.1 (Representation)**.** A *(linear) representation* of a group G on a finite-dimensional complex vector space V is a homomorphism

$$\rho\colon G \longrightarrow \mathrm{GL}(V),$$

where $\mathrm{GL}(V)$ is the group of invertible linear operators on V. Thus to each $g \in G$ is assigned an invertible operator $\rho(g)$, with

$$\rho(gh) = \rho(g)\,\rho(h) \quad \text{for all } g, h \in G,$$

whence automatically $\rho(e) = \mathrm{id}_V$ and $\rho(g^{-1}) = \rho(g)^{-1}$. The *dimension* (or degree) of the representation is $\dim V$. We speak of "the representation (ρ, V)", or simply of V when ρ is understood.

Unless stated otherwise our vector spaces are complex and finite-dimensional; the real case will be flagged when it matters (and it will matter — see Example 3.13).

Notation: operators versus matrices. Throughout these notes we keep two parallel notations, and the distinction is deliberate. The symbol $\rho(g)$ always denotes the abstract

linear *operator* on V, defined without reference to any basis. Once a basis $(e_1, \dots, e_n)$ of V is chosen, the corresponding *matrices* are denoted $D(g)$ and defined by

$$\rho(g)\, e_j = \sum_{i=1}^{n} D_{ij}(g)\, e_i .$$

The j-th column of $D(g)$ lists the components of the image of the j-th basis vector. The homomorphism property becomes matrix multiplication, $D(gh) = D(g)D(h)$, and a map $D\colon G \to \mathrm{GL}(n,\mathbb{C})$ with this property is called a *matrix representation*. Irreducible representations, once we have them, will carry labels $\rho^{(i)}$, $D^{(i)}$. Keeping operators and matrices typographically apart costs little and prevents a great deal of confusion when several bases are in play at once — as they constantly are from Lecture 4 onwards.

Example 3.2 (One-dimensional representations)**.** On a one-dimensional space, an invertible operator is multiplication by a nonzero number: $\mathrm{GL}(1,\mathbb{C}) = \mathbb{C}\setminus\{0\}$. A one-dimensional representation is therefore nothing but a homomorphism $G \to \mathbb{C}\setminus\{0\}$, and we have already met several:

(i) the *trivial representation* $\rho(g) = 1$ for all g, defined for every group, on which symmetry acts invisibly;

(ii) the *sign representation* $\mathrm{sgn}\colon S_n \to \{\pm 1\}$ of Lecture 1;

(iii) the *orientation representation* $\nu\colon D_n \to \{\pm 1\}$ of Lecture 2, with $\nu = +1$ on rotations and -1 on reflections. ◁

Example 3.3 (All the χ_k of a cyclic group)**.** Let $C_n = \langle r \rangle$ and $\omega = e^{2\pi \mathrm{i}/n}$. For each $k \in \{0, 1, \dots, n-1\}$,

$$\chi_k(r^m) = \omega^{km} = e^{2\pi \mathrm{i} km/n}$$

is a one-dimensional representation: $\chi_k(r^a)\chi_k(r^b) = \omega^{k(a+b)} = \chi_k(r^{a+b})$, and the value is well defined because $\omega^{kn} = 1$. These n representations are pairwise distinct ($\chi_k(r) = \omega^k$ differ), and in Lecture 4 we shall see that they are *all* the irreducible representations of C_n — the finite Fourier modes, as we will recognise when characters appear. ◁

Example 3.4 (The defining representation of D_n)**.** Realising D_n by orthogonal transformations of the plane, as in Lecture 1, *is* a two-dimensional real representation, called the *defining* (or vector) representation. In the basis with vertex 1 on the y-axis (Sheet 1 instead puts the mirror of s on the x-axis, $s = \mathrm{diag}(1,-1)$; both choices satisfy the dihedral relations and are conjugate by a rotation through $\pi/2$), the generators of D_3 have matrices

$$D(r) = \begin{pmatrix} \cos\frac{2\pi}{3} & -\sin\frac{2\pi}{3} \\ \sin\frac{2\pi}{3} & \cos\frac{2\pi}{3} \end{pmatrix} = \frac{1}{2}\begin{pmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix}, \qquad D(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Multiplying the matrices out, we check that

$$D(r)^3 = D(s)^2 = \mathbb{1}, \qquad D(s)D(r)D(s)^{-1} = D(r)^{-1},$$

i.e. the dihedral relations hold. That is enough to define D on the whole group. Lecture 1 listed the elements of D_n as $r^a s^\varepsilon$ with $a \in \{0,1,\dots,n-1\}$ and $\varepsilon \in \{0,1\}$, and each element appears in that list exactly once. We call this way of writing an element its *normal form*. Set

$$D(r^a s^\varepsilon) = D(r)^a D(s)^\varepsilon .$$

Because the normal form is unique, this formula assigns exactly one matrix to each element, so D is well defined.

The same element can also be written in other ways. In D_3, for instance, $sr = r^2 s$. The formula reads only normal forms, so it says nothing directly about the product $D(s)D(r)$. The relations show that this product is the right matrix:

$$D(s)D(r) = D(r)^{-1}D(s) = D(r)^2 D(s) = D(r^2 s).$$

The same happens for every product. D is multiplicative, because the same three relations that give Lecture 1's product rule $(r^a s^\varepsilon)(r^b s^\delta) = r^{a+(-1)^\varepsilon b} s^{\varepsilon+\delta}$ hold for the matrices. ◁

Groups often arrive acting on a *set* before they act on a vector space; the following definition names what we have in fact been doing since Lecture 1.

Definition 3.5 (Group action)**.** An *action* of a group G on a set X is a map

$$G \times X \to X, \quad (g, x) \mapsto g \cdot x,$$

such that $e \cdot x = x$ and $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$, $x \in X$.

Examples of actions: symmetry operations acting on the points of space; S_n acting on $\{1, \dots, n\}$; and G acting on itself by conjugation (Lecture 2).

Example 3.6 (Permutation representations)**.** Let G act on a finite set X. Take a vector space with one basis vector e_x for each point $x \in X$, namely $V = \mathbb{C}^X$, and let G permute the basis the way it permutes the points:

$$\rho(g)\, e_x = e_{g \cdot x}.$$

Then $\rho(g)\rho(h)e_x = e_{g\cdot(h\cdot x)} = e_{(gh)\cdot x} = \rho(gh)e_x$: a representation of dimension $|X|$, the *permutation representation* of the action. Its matrices are *permutation matrices* — a single 1 in each row and column. For S_3 acting on $\{1, 2, 3\}$, the generators $(1\,2)$ and $(1\,2\,3)$ act on the basis (e_1, e_2, e_3) by

$$D\big((1\,2)\big) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad D\big((1\,2\,3)\big) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

the columns recording $e_1 \mapsto e_2$, $e_2 \mapsto e_3$, $e_3 \mapsto e_1$ in the second case. This representation is decomposed in full in Lecture 4. Taking $X = G$ with the action of G on itself by left multiplication yields the *regular representation*, of dimension $|G|$, which returns in Lecture 5. ◁

Remark 3.7 (Faithfulness)**.** A representation is *faithful* if ρ is injective ($\ker \rho = \{e\}$), i.e. if the matrices remember the whole group. The defining representation of D_n and all permutation representations of group actions in which only e fixes every point (actions that *separate points*) are faithful; the trivial representation is maximally unfaithful. There is no requirement of faithfulness in the theory: a representation may legitimately see only a quotient $G/\ker\rho$ — recall the orientation representation, which sees only orientation.

3.2 How symmetries act on functions

We now come to the single most important construction of the course. The states of physical systems are very often *functions* on the space the symmetry group transforms: wavefunctions

on configuration space, displacement fields on a molecule, temperature distributions on a body. The group's action on points then induces an action on the functions — and it does so by a formula with one subtle feature: an inverse.

Theorem 3.8 (The function-space representation)**.** Let the group G act on a set X, and let $\mathcal{F}(X)$ denote the vector space of complex-valued functions on X. For $g \in G$ and $\psi \in \mathcal{F}(X)$ define

$$\big(\rho(g)\psi\big)(x) := \psi\big(g^{-1} \cdot x\big).$$

Then each $\rho(g)$ is an invertible linear operator and $\rho\colon G \to \mathrm{GL}(\mathcal{F}(X))$ is a representation. The same formula defines a representation on any subspace of $\mathcal{F}(X)$ stable under all the $\rho(g)$ (for infinite X one works with such subspaces — polynomials, smooth functions, square-integrable functions — as the context requires).

Proof. Linearity in ψ is clear, and $\rho(e) = \mathrm{id}$. The homomorphism property is the following computation:

$$\big(\rho(g)\rho(h)\psi\big)(x) = \big(\rho(h)\psi\big)(g^{-1} \cdot x) = \psi\big(h^{-1} \cdot (g^{-1} \cdot x)\big) = \psi\big((gh)^{-1} \cdot x\big) = \big(\rho(gh)\psi\big)(x),$$

using $h^{-1}g^{-1} = (gh)^{-1}$ from Lecture 1. Invertibility follows: $\rho(g)\rho(g^{-1}) = \rho(e) = \mathrm{id}$. ∎

Why the inverse? First the conceptual answer: $\rho(g)\psi$ is the function ψ *carried along by* g. Evaluating the definition at a transformed point,

$$\big(\rho(g)\psi\big)(g \cdot x) = \psi(x).$$

Read it like this: the transformed function $\rho(g)\psi$ takes, at the transported point $g \cdot x$, the value that the original function ψ took at the original point x.

For example, let X be finite, and let δ_x be the function equal to 1 at the point x and to 0 elsewhere. Then $\big(\rho(g)\delta_x\big)(y) = \delta_x(g^{-1} \cdot y)$ equals 1 exactly when $g^{-1} \cdot y = x$, that is, when $y = g \cdot x$. Hence

$$\rho(g)\,\delta_x = \delta_{g \cdot x}.$$

The spike that sat at x now sits at $g \cdot x$: it moves together with the point. The functions δ_x play the role of the basis vectors e_x of Example 3.6.

Second, the mechanical answer: the inverse is forced by the order of composition. Had we defined $(\sigma(g)\psi)(x) = \psi(g \cdot x)$, the same computation would give $\sigma(g)\sigma(h) = \sigma(hg)$ — the products in the wrong order, an *anti*-homomorphism. The inverse undoes that reversal.

The action of permutations on polynomials, mentioned at the end of Lecture 1, is the same mechanism written in coordinate slots. Take the left action

$$\sigma \cdot (x_1, \dots, x_n) = (x_{\sigma^{-1}(1)}, \dots, x_{\sigma^{-1}(n)}).$$

It is an action, because the i-th entry of $\sigma \cdot (\tau \cdot x)$ is $x_{\tau^{-1}\sigma^{-1}(i)} = x_{(\sigma\tau)^{-1}(i)}$. Its inverse sends x to $(x_{\sigma(1)}, \dots, x_{\sigma(n)})$, so

$$(P_\sigma f)(x_1, \dots, x_n) = f(x_{\sigma(1)}, \dots, x_{\sigma(n)}).$$

In words, P_σ replaces each variable x_i by $x_{\sigma(i)}$. By Theorem 3.8 this is a representation, and the composition can also be checked directly:

$$\begin{aligned}\big(P_\sigma(P_\tau f)\big)(x_1, \dots, x_n) &= (P_\tau f)(x_{\sigma(1)}, \dots, x_{\sigma(n)}) \\ &= f(x_{\sigma(\tau(1))}, \dots, x_{\sigma(\tau(n))}) = (P_{\sigma\tau} f)(x_1, \dots, x_n).\end{aligned}$$

The tempting alternative replaces x_i by $x_{\sigma^{-1}(i)}$. Call it Q_σ. The same computation gives $Q_\sigma Q_\tau = Q_{\tau\sigma}$: the product comes out in the wrong order, exactly as for the rule without the inverse above.

Example 3.9 (The functions x and y)**.** Let D_3 act on the plane as in Example 3.4, with the vertices of the triangle at

$$\vec{p}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad \vec{p}_2 = \begin{pmatrix} -\sqrt{3}/2 \\ -1/2 \end{pmatrix}, \qquad \vec{p}_3 = \begin{pmatrix} \sqrt{3}/2 \\ -1/2 \end{pmatrix},$$

so that r carries $\vec{p}_1 \mapsto \vec{p}_2 \mapsto \vec{p}_3 \mapsto \vec{p}_1$. Consider the two coordinate functions x and y. To find $\rho(r)x$, we need r^{-1}, the rotation through $-2\pi/3$:

$$r^{-1}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -\frac{1}{2}x + \frac{\sqrt{3}}{2}y \\ -\frac{\sqrt{3}}{2}x - \frac{1}{2}y \end{pmatrix}.$$

The function $\rho(r)x$ takes at the point (x, y) the value of x at $r^{-1}(x, y)$, which is the first component. Likewise for y; and s is its own inverse, $s(x, y) = (-x, y)$. So

$$\rho(r)x = -\tfrac{1}{2}x + \tfrac{\sqrt{3}}{2}y, \qquad \rho(r)y = -\tfrac{\sqrt{3}}{2}x - \tfrac{1}{2}y, \qquad \rho(s)x = -x, \qquad \rho(s)y = y.$$

The coefficients of $\rho(r)x$ form the first column of the matrix of $\rho(r)$ in the basis (x, y), and those of $\rho(r)y$ the second:

$$\begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} = D(r), \qquad \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = D(s).$$

The coordinate functions carry exactly the defining representation. As a check on the inverse, evaluate $\rho(r)x$ at the vertex $\vec{p}_1$: $-\frac{1}{2}\cdot 0 + \frac{\sqrt{3}}{2}\cdot 1 = \frac{\sqrt{3}}{2}$, which is the value of x at $\vec{p}_3$, the vertex that r carries to $\vec{p}_1$. ◁

This is physics, not only bookkeeping. The $2p_x$ and $2p_y$ orbitals of an atom have the form $x\, f(|\vec{r}|)$ and $y\, f(|\vec{r}|)$, with the same radial factor. For an atom at the centre of the triangle the symmetries leave $|\vec{r}|$ unchanged, so the two orbitals transform like x and y. In the basis $(2p_x, 2p_y)$ the matrix of $\rho(g)$ is therefore the same $D(g)$ as in the basis (x, y). In Lecture 8 the same fact tells us how an electric dipole transforms. Higher-degree polynomials furnish further representations — the quadratics x^2, xy, y^2 carry a three-dimensional one — whose decomposition is a short computation once we have the tools of Lecture 5.

[self-study] *The same for any group of orthogonal matrices.* Let $G \subseteq \mathrm{O}(2)$ act on the plane, and write x_1, x_2 for the coordinate functions, $x_i(\vec{r}\,) = r_i$. Then

$$\big(\rho(g)x_i\big)(\vec{r}\,) = x_i\big(g^{-1}\vec{r}\,\big) = (g^{-1})_{ij}\, r_j = (g^{\mathsf{T}})_{ij}\, r_j = g_{ji}\, x_j(\vec{r}\,),$$

summation over j understood and orthogonality $g^{-1} = g^{\mathsf{T}}$ used. Hence $\rho(g)x_i = \sum_j g_{ji}x_j$: comparing with the convention $\rho(g)e_j = \sum_i D_{ij}(g)e_i$, the matrix of $\rho(g)$ in the basis (x_1, x_2) is g itself.

The dual representation *[self-study]*

For a general (not necessarily orthogonal) representation, the computation of the self-study paragraph above does not return the same matrices, and what it does return deserves its own name. Recall that the *dual space* V^* of V is the space of linear maps $\varphi : V \to \mathbb{C}$ (*functionals*); given a basis (e_i) of V, the dual basis (e^i) of V^* is defined by $e^i(e_j) = \delta_{ij}$. Linear functionals are functions on V, so the function-space construction applies to them, and it preserves linearity:

Proposition 3.10 (Dual representation)**.** Let (ρ, V) be a representation. Then

$$\big(\rho^*(g)\varphi\big)(v) := \varphi\big(\rho(g^{-1})v\big), \qquad \varphi \in V^*,\ v \in V,$$

defines a representation (ρ^*, V^*), the *dual* (or *contragredient*) representation. In the dual basis its matrices are

$$D^*(g) = D(g^{-1})^{\mathsf{T}} = \big(D(g)^{-1}\big)^{\mathsf{T}}.$$

Proof. Each $\rho^*(g)$ maps functionals to functionals linearly, and the homomorphism property is Theorem 3.8 applied to the action of G on the set V via ρ. For the matrices: $\big(\rho^*(g)e^j\big)(e_i) = e^j\big(\rho(g^{-1})e_i\big) = D_{ji}(g^{-1})$, so expanding $\rho^*(g)e^j$ in the dual basis gives coefficients $D^*_{ij}(g) = D_{ji}(g^{-1})$, which is the stated matrix. One can also verify directly that $g \mapsto D(g^{-1})^{\mathsf{T}}$ is a homomorphism: transposition and inversion each reverse the order of products, and two reversals cancel. ∎

If the matrices $D(g)$ are *unitary* — and for a *finite* group Lecture 4 shows this can always be arranged — then $D(g^{-1}) = D(g)^\dagger$ and $D^*(g) = \overline{D(g)}$: *the dual representation is the complex conjugate representation.* The physics reader has met this contragredient behaviour before: if kets transform as $|\psi\rangle \mapsto \rho(g)\,|\psi\rangle$, then bras transform with the conjugate matrices so that $\langle\varphi|\psi\rangle$ stays invariant. The distinction between a representation and its conjugate, invisible when the matrices are real, prepares an important distinction in later SU(3) study: the quark triplet $\mathbf{3}$ and the antiquark antitriplet $\bar{\mathbf{3}}$ are conjugate representations. This course builds the dual/conjugate vocabulary but does not teach that particle-physics application; Jones, Ch. 8, §§8.3–8.4 is a further-reading destination.

The function-space construction is fully general, and we shall lean on it repeatedly. It is how a finite group acts on polynomials (the sign of a permutation, Lecture 1, and Example 3.9); how a point group acts on the displacement patterns of a vibrating molecule (Lecture 7); how a symmetry group acts on quantum wavefunctions, $\psi(\vec{r}\,) \mapsto \psi(g^{-1}\vec{r}\,)$, which is the formula underlying degeneracies and selection rules (Lectures 6 and 8); and how the rotation group acts on functions on the sphere, whose stable subspaces are spanned by the spherical harmonics (Part II). The same formula serves in every one of these places.

3.3 Equivalence of representations

When are two representations "the same"? As with groups themselves (Lecture 1), the answer is: when a structure-preserving dictionary translates one into the other.

Definition 3.11 (Equivalence)**.** Representations (ρ_V, V) and (ρ_W, W) of the same group G are *equivalent*, written $\rho_V \cong \rho_W$, if there is an invertible linear map $T : V \to W$ with

$$T\,\rho_V(g) = \rho_W(g)\,T \quad \text{for all } g \in G.$$

Such a T is called an *intertwiner* (it lets the group action pass through it).

In matrix language: choosing bases and letting S be the matrix of T, equivalence reads $D_W(g) = S\,D_V(g)\,S^{-1}$ for all g, with one and the same S. In particular the matrices of a single representation in two different bases are related this way — equivalent representations are

one representation written in different coordinates, and all notions we define (irreducibility, characters, multiplicities) will be invariant under equivalence.

Two representations can be combined into one bigger representation.

Definition 3.12 (Direct sum)**.** Let (ρ, V) and (σ, W) be representations of G. The space $V \oplus W$ consists of all pairs (v, w) with $v \in V$ and $w \in W$. The *direct sum* $\rho \oplus \sigma$ of the two representations acts on it by

$$(\rho \oplus \sigma)(g)\,(v, w) = \big(\rho(g)v,\ \sigma(g)w\big).$$

Each component of the pair is transformed on its own, so $\rho \oplus \sigma$ is again a representation. In a basis of V followed by a basis of W, its matrices are block-diagonal:

$$D_{V\oplus W}(g) = \begin{pmatrix} D_V(g) & 0 \\ 0 & D_W(g) \end{pmatrix}.$$

Example 3.13 (A representation that splits over $\mathbb{C}$ but not over $\mathbb{R}$)**.** Let $C_3 = \langle r \rangle$ act on $\mathbb{C}^2$ through the rotation matrices of Example 3.4: $D(r^k) = R(2\pi k/3)$. The vectors $v_\pm = \left(\begin{smallmatrix} 1 \\ \mp \mathrm{i} \end{smallmatrix}\right)$ are eigenvectors of every rotation:

$$R(\theta)\begin{pmatrix} 1 \\ \mp\mathrm{i} \end{pmatrix} = \begin{pmatrix} \cos\theta \pm \mathrm{i}\sin\theta \\ \sin\theta \mp \mathrm{i}\cos\theta \end{pmatrix} = e^{\pm \mathrm{i}\theta}\begin{pmatrix} 1 \\ \mp\mathrm{i} \end{pmatrix},$$

as multiplying out confirms. So in the basis (v_+, v_-) the matrices become

$$S^{-1}D(r^k)\,S = \begin{pmatrix} \omega^k & 0 \\ 0 & \omega^{-k} \end{pmatrix}, \qquad \omega = e^{2\pi\mathrm{i}/3},$$

where $S = (v_+ \mid v_-)$ is the matrix whose columns are the vectors v_+ and v_-. So the rotation representation of C_3 is equivalent, over $\mathbb{C}$, to the direct sum $\chi_1 \oplus \chi_2$ (indices mod 3, so $\chi_2 = \chi_{-1}$) of two of the one-dimensional representations of Example 3.3. The diagonal matrix above has the 1×1 blocks $\chi_1(r^k) = \omega^k$ and $\chi_2(r^k) = \omega^{2k} = \omega^{-k}$. Notice that the diagonalising basis is genuinely complex: over the *reals* a rotation through $2\pi/3$ has no eigenvector at all, and the representation cannot be split. Working over $\mathbb{C}$ is not a convenience but a structural choice, and it is the right one for quantum mechanics, where state spaces are complex from the outset. ◁

The equivalence just found does two things at once: it relates two ways of writing the same representation, and it splits the rotation representation into two smaller ones. Splitting is the theme of the next lecture. It asks when a representation contains a smaller one, whether the smaller one always comes with a complement, and what the pieces that cannot be split further look like.

Summary.

- *Representation.* A homomorphism $G \to \mathrm{GL}(V)$, equivalently a linear action of G on V (Definition 3.1); in a basis, a matrix representation $D(gh) = D(g)D(h)$. First examples: the one-dimensional χ_k of C_n, the defining representation of D_n, permutation representations.
- *Function space.* G acts on functions by $(\rho(g)\psi)(x) = \psi(g^{-1}x)$ (Theorem 3.8); the inverse keeps the order of products. The self-study section carries the dual representation.

- *Equivalence.* Two representations are the same when an invertible intertwiner T with $T\rho_V(g) = \rho_W(g)T$ connects them (Definition 3.11). Over $\mathbb{C}$ the rotation representation of C_3 is equivalent to the direct sum $\chi_1 \oplus \chi_2$ (Definition 3.12); over $\mathbb{R}$ it cannot be split.

Tutorial. Exercise Sheet 3.

References

Jones, *Groups, Representations and Physics*, Ch. 3; Tung, *Group Theory in Physics*, §§3.1–3.2; Fulton–Harris, *Representation Theory*, §1.1; Serre, *Linear Representations of Finite Groups*, §§1.1–1.3.